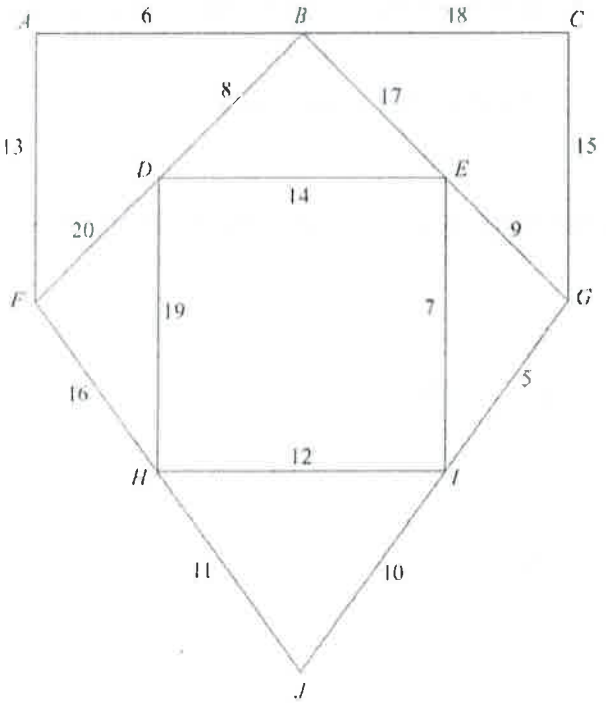
Name: _____ **Branch:** _____ **SNU ID:** _____

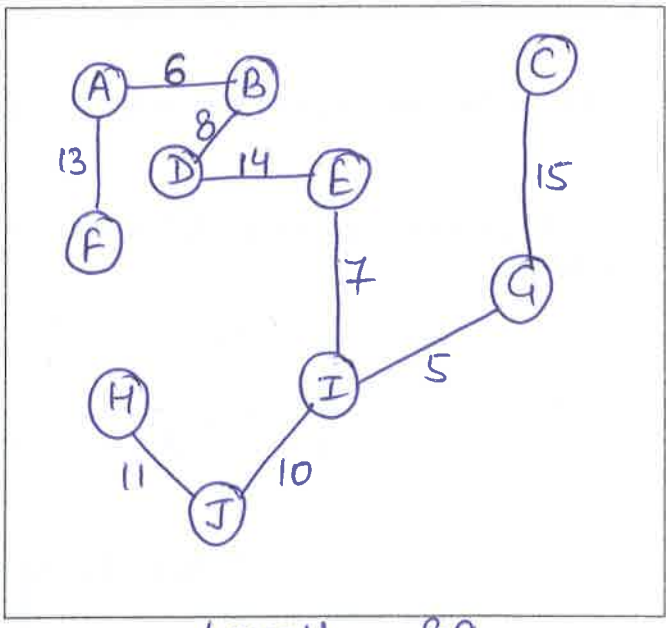
Note:- **Q.3, Q.7, & Q.8 – 3 Marks each, Q.10 – 2 Marks, Remaining Questions – 1 Mark each**

1. Which of the following ways can be used to represent a graph?
   (i) Adjacency List and Adjacency Matrix
   (ii) Adjacency List, Adjacency Matrix as well as Incidence Matrix
   (iii) Incidence Matrix
   (iv) None of the mentioned

2. Which of the following is an advantage of adjacency list representation over adjacency matrix representation of a graph?
   (i) In adjacency list representation, space is saved for sparse graphs.
   (ii) DFS and BSF can be done in O(V + E) time for adjacency list representation. These operations take $O(V^2)$ time in adjacency matrix representation. Here is V and E are number of vertices and edges respectively.
   (iii) Adding a vertex in adjacency list representation is easier than adjacency matrix representation.
   (iv) All of the above

3. The following graph has 10 vertices: A, B, ….., J. The numbers on each edge represent the distances, in miles, between pairs of vertices.





Length = 89

Use Kruskal's Algorithm to find MST for the graph and state the length of your spanning tree. Draw the spanning tree as well.

4. Which one of the following algorithm solves the positive weighted single source shortest path problem?
   (i) Breadth First Search
   (ii) Depth First Search
   (iii) Kruskal's Algorithm
   (iv) Dijkstra's Algorithm

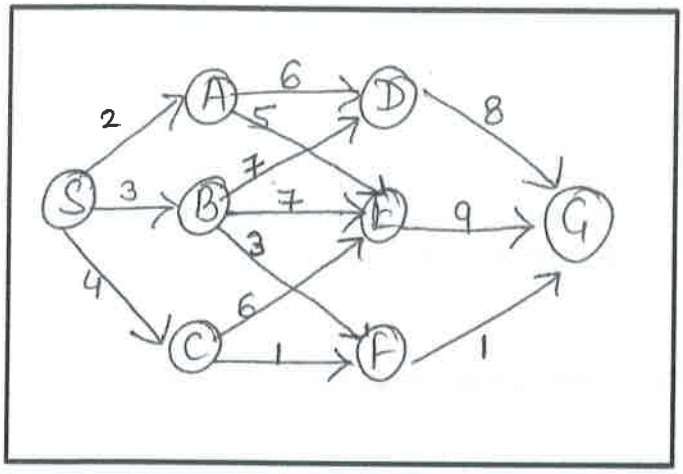
5. **Consider the given statements:**
   I. Uses priority queue to store unvisited vertices by distance from source.
   II. It uses greedy method, means pick the next closest vertex to the source.
   III. Does not work with negative weights.

   The above statements describe
   (i) Bellman Ford Algorithm
   (ii) Dijkstra's Algorithm
   (iii) Kruskal's Algorithm
   (iv) Floyd Warshall Algorithm

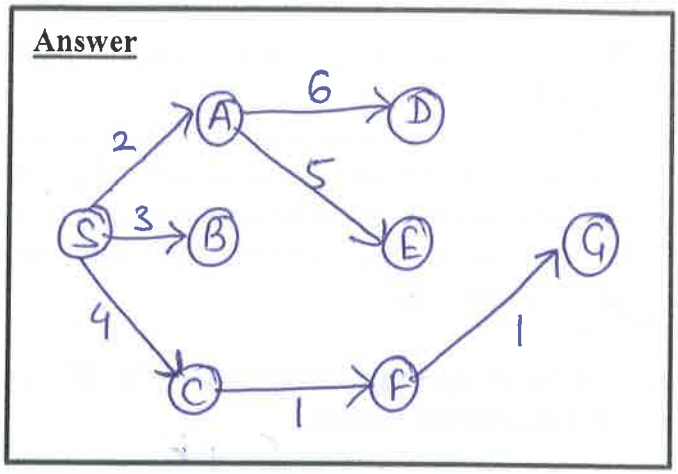
6. The Bellman Ford Algorithm solves the single-source shortest path problem in the case in which edge weights may be negative, what is the time complexity of running Bellman Ford Algorithm?

(i) $O(V^2)$ (ii) O(V * E) (iii) O(V + E) (iv) O(E log V)

7. **Consider the given multi stage graph**



Answer

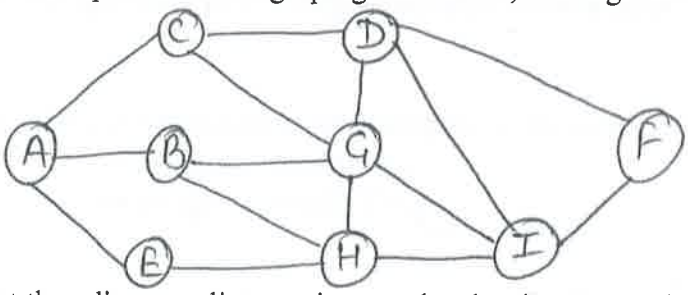


What is the shortest path from node 'S' to node 'G' using Dijkstra's approach?

8. Consider the same graph given in Q7, what is the shortest path from node 'S' to node 'G' using Floyd Warshall Algorithm?

Total length = 6

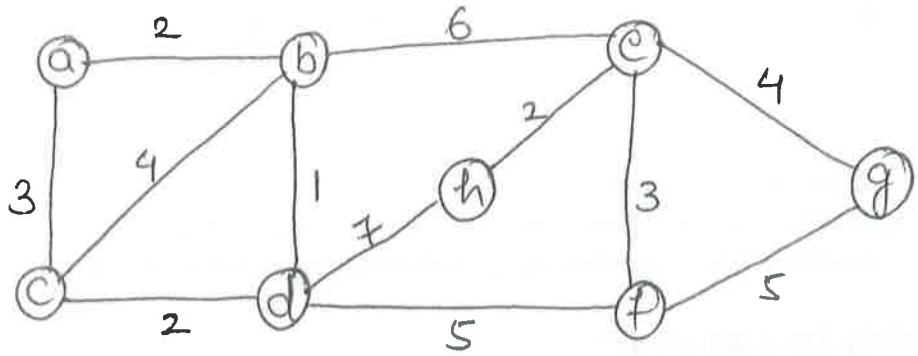
9. Write the BFS sequence on the graph given below, starting at vertex A.



Assume that the adjacency lists are in sorted order. For Example, when exploring vertex E, the algorithm considers the edge E-B, before E-C, E-F, E-G, or E-H. Which of the following is the order of vertices that are enqueued on the FIFO queue?

(i) A, B, C, E, G, H, I, D, F
(ii) A, B, C, E, H, G, D, I, F
(iii) A, B, C, D, E, G, H, I, F
(iv) A, B, C, E, G, H, D, I, F

10. Consider the given graph



Which of the following can be the sequence of edges added to minimum spanning tree using 'Prim's Algorithm

(i) (a-b), (b-d), (d-c), (d-f), (f-g), (f-e), (e-h)
(ii) (a-b), (b-d), (d-c), (d-f), (f-e), (e-g), (e-h)
(iii) (e-h), (e-f), (e-g), (f-d), (d-b), (b-a), (d-c)
(iv) (e-h), (e-f), (e-g), (d-b), (f-d), (b-a), (d-c)

11. The distance matrix of a graph with vertices A, B, C and D is given below:

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0 & 1 & \infty & \infty \\ \infty & 0 & 2 & 4 \\ 3 & \infty & 0 & 1 \\ 1 & \infty & \infty & 0 \end{bmatrix} \end{array}$$

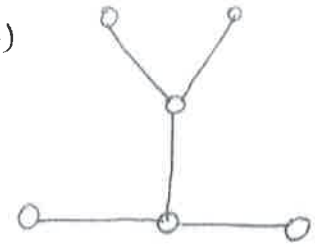
The shortest path from B to D consists of edge(s) _____.

(i) BC and CD
(ii) BD
(iii) AB and CD
(iv) There is no shortest path

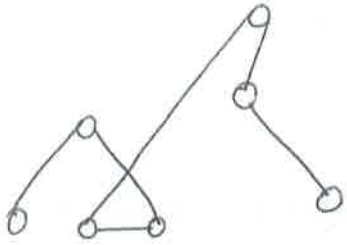
(iv) None of these

12. Which of the following is not a tree?

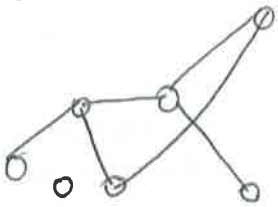
(i)



(ii)



(iii)



13. What is the number of vertices in a spanning tree with 57 edges?

(i) 58 (ii) $2^6 - 4$ (iii) 56 (iv) 57

14. The reason for using Bellman-Ford algorithm for finding the shortest path between 2 vertices in a graph, when Dijkstra's algorithm does the same thing:

I. Bellman-Ford algorithm is faster than Dijkstra's Algorithm
II. Bellman-Ford Algorithm works on any directed graph, with negative weights also.

Which of the following is true?

(i) I only
(ii) II only
(iii) Both I and II
(iv) None of these

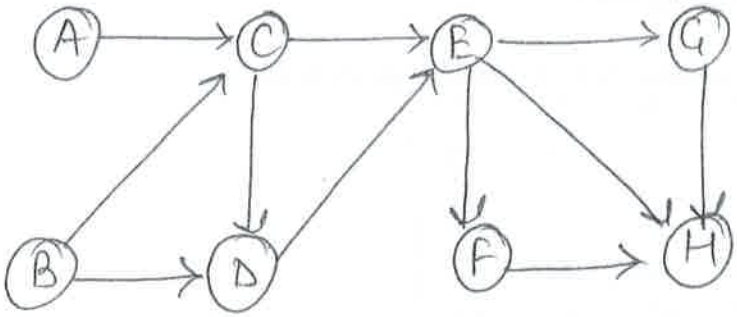
15. If we run Dijkstr's algorithm starting from 'S' to find shortest path 'T', consider the following statemnets:

I. Dijkstra's algorithm returns shortest path with minimum total weight.
II. Dijkstra's algorithm returns shortest path, with minimum number of edges.

Which of the following is true?

(i) I only
(ii) II only
(iii) Both I and II
(iv) None of these

16. Consider the following graph:



Which of the following orderings not a valid topological sort of the graph?

(i) BACEDFGH
(ii) ABCDEGFH
(iii) ABCDEFGH
(iv) BACDEFGH

17. Consider an undirected graph

G = (V, E)
V = {r, s, t, u, v, w, x, y }
E = { (r, s), (r, v), (s, w), (w, t), (w, x), (t, u), (t, x), (u, y), (u, x), (x, y) }

Which node has the highest degree?

(i) *w* (ii) *x* (iii) *u* (iv) *t*